Comprehensive Technical Evaluation, QA & QC Audit Report

Target System: Nueva Ecija Travel Itinerary Application ("Trip mo, trip ko") • Repository Commit: 06cefc5 (v0.15.0-rc.6) • Evaluation Date: November 2, 2026

Audit Parameter	Verified Technical Value
GitHub Repository	https://github.com/Edselrenz/Itinerary-App (Default branch: `main`, 53 commits, 0 tags)
Deployed GitHub Pages URL	https://edselrenz.github.io/Itinerary-App/ (SPA with `404.html` fallback & `.nojekyll`)
Evaluated Commit & Version	Commit `06cefc5997cdc07ce29b5b6a4f2db18d146b0498` (`06cefc5`) • Release `v0.15.0-rc.6` (November 2, 2026)
Frontend Technology Stack	Angular 22.2.3 (Standalone Components), Ionic Framework 9, Capacitor 8.6.0, Leaflet 1.9.4 + MarkerCluster 1.5.3, RxJS 7.8.0
Backend Technology Stack	Python 3.11+, FastAPI 0.1.0, SQLAlchemy 2.0 + Alembic, PostgreSQL, SlowAPI Rate Limiter, Pydantic Settings
Live Backend Tunnel Endpoint	`https://anybody-cube-surprise.ngrok-free.dev/api/v1` (configured in `Project/src/environments/environment.prod.ts`)
Final Verdict	**NEEDS MINOR IMPROVEMENTS** — Generally functional with resilient offline/online dual-mode architecture, but contains verified implementation, contract, and UX defects requiring remediation before production sign-off.

A. Executive Summary

This report presents an evidence-driven Senior Software Engineering, Code Audit, Quality Assurance (QA), and Quality Control (QC) evaluation of the Edselrenz/Itinerary-App monorepo and its live GitHub Pages deployment at https://edselrenz.github.io/Itinerary-App/. The evaluation encompassed static inspection of the frontend and backend source trees across 53 commits (culminating in commit `06cefc5`, version `0.15.0-rc.6`) alongside active interactive testing of the deployed application connected to its live FastAPI backend tunnel and in its offline fallback state.

1. Overall Project & Functional Condition

The Nueva Ecija Travel Itinerary Application ("Trip mo, trip ko") is a hybrid web and mobile travel discovery and trip-planning platform scoped to Nueva Ecija, Central Luzon, Philippines. Unlike typical academic prototypes that fail completely when their backend is unreachable, this application implements a dual-mode resilience strategy: when the FastAPI backend is reachable via the configured ngrok tunnel (`https://anybody-cube-surprise.ngrok-free.dev/api/v1`), the Angular client fetches live Points of Interest (POIs) and generates Haversine-clustered itineraries from PostgreSQL; when the backend tunnel is unreachable or the user operates in unauthenticated Guest Mode, `PlaceService`, `ItineraryService`, and `GuestStorageService` transparently degrade to local IndexedDB/LocalStorage persistence and embedded Nueva Ecija place catalogs.

Core user journeys—including splash screen dismissal, Guest Mode onboarding, Home destination search and category filtering, Place Detail inspection, Leaflet interactive map filtering, 3-step Suggested Itinerary generation, Custom Itinerary timeline building with stop reordering, and local itinerary persistence across browser refreshes—were verified working in live browser execution. However, several secondary workflows and backend code paths exhibit concrete defects, including stubbed password-reset and Google OAuth handlers that display misleading success/status toasts without network execution, duplicate route registrations in the FastAPI itinerary router causing dead code and missing timestamp updates, triple-duplicated offline place catalogs across three frontend TypeScript files, and documentation drift between `docs/KNOWN_ISSUES.md` and the actual UI.

Evaluation Dimension	Assessed Condition	Key Evidence & Summary Assessment
General Code Quality	Good with Maintainability Debt	Strict TypeScript and Pydantic schemas are used throughout, and 16 prior audit items (`CR-001` to `CR-016`) were remediated. However, `itinerary.service.ts` (1,007 lines), `custom-builder.page.ts` (1,003 lines), and `itineraries.py` (706 lines) are monolithic and embed duplicated POI datasets.
Functional Condition	Core Working / Secondary Stubbed	Guest exploration, Home search/filtering, Map clustering, Suggested Wizard (online + offline), and Custom Builder CRUD work reliably. Forgot Password, Google Sign-In/Sign-Up, and Apple Sign-In on `/login` and `/signup` are non-functional UI stubs.
QA Findings	30 Executed & Documented Cases	21 Passed, 4 Failed, 3 Partially Passed, 2 Not Tested (require authenticated production OAuth or multi-user live DB state).
QC Findings	Solid Architecture / Data Duplication	Clean separation between Angular core services and feature pages, but offline POI data is duplicated in 3 separate frontend files (`home.page.ts`, `place.service.ts`, `itinerary.service.ts`) with inconsistent schemas and counts.
Deployment Condition	Verified Consistent (Commit `06cefc5`)	GitHub Actions (`.github/workflows/deploy.yml`) builds `Project/` with `--base-href /Itinerary-App/` and copies `index.html` to `404.html` for SPA deep-link routing. Commit `06cefc5` removed the legacy `[Set Server]` prompt and baked in the ngrok endpoint with `ngrok-skip-browser-warning: true`.
Security Condition	Strong Baseline / Minor Gaps	BOLA/IDOR checks (`Itinerary.user_id == current_user.id`), SQL LIKE escaping, SlowAPI rate limiting, and HTML escaping in map helpers are present. Remaining concerns include JWT storage in `localStorage`, placeholder Google Client ID in `environment.prod.ts`, and unguarded `localStorage` custom URL override without domain allowlisting.

B. Repository Analysis (Phase 1 Reconnaissance)

1. Architecture & Monorepo Organization

The repository (Edselrenz/Itinerary-App) is structured as a single-branch (`main`) full-stack monorepo containing four primary top-level domains:

- **Frontend Application (`Project/`):** An Ionic 9 + Angular 22 standalone-component Single-Page Application (SPA) and Capacitor 8 hybrid mobile project. It follows a layered feature module layout under `Project/src/app/` divided into `core/` (`interceptors/`, `models/`, `services/`, `utils/`), `features/` (`auth/` and `itinerary/`), and `tabs/` (`home/`, `map/`, `history/`, `profile/`).
- **Backend API (`backend/`):** A modular Python FastAPI REST service organized into `app/api/v1/` (route controllers for `auth.py`, `places.py`, and `itineraries.py`), `app/algorithms/` (deterministic Haversine clustering generator in `itinerary_generator.py`), `app/core/` (`config.py`, `database.py`, `limiter.py`), `app/models/` (SQLAlchemy 2.0 ORM entities in `entities.py`), `app/schemas/` (Pydantic v2 request/response contracts), and `app/services/` (`auth_service.py`). Database migrations are managed via Alembic under `backend/alembic/`.
- **Automation & Seeding Scripts (`scripts/`):** Operational utilities including `setup_postgres.py`, `seed_featured_places.py`, `import_osm.py`, and `db_backup_restore.py`.
- **Engineering Documentation (`docs/`):** Seventeen Markdown documents tracking architecture (`SYSTEM_ARCHITECTURE.md`), API contracts (`API_DOCUMENTATION.md`), audit history (`CODE_REVIEW_AUDIT.md`), known limitations (`KNOWN_ISSUES.md`), and session logs (`DEVELOPMENT_LOG.md`, `CHANGELOG.md`).

2. Routing, State Management & Data Persistence

Frontend routing is defined in Project/src/app/app.routes.ts and Project/src/app/tabs/tabs.routes.ts using lazy-loaded standalone components (`loadComponent` / `loadChildren`) with `PreloadAllModules` and `withComponentInputBinding()` enabled in Project/src/main.ts. Top-level routes comprise `/welcome`, `/login`, `/signup`, `/guest`, `/custom-builder`, `/custom-builder/:id`, `/suggested-wizard`, `/itinerary/:id`, and `/tabs` (with child routes `/tabs/home`, `/tabs/map`, `/tabs/history`, and `/tabs/profile`). Unmatched routes (`**`) redirect to `/welcome`.

State management relies on RxJS `BehaviorSubject` streams inside singleton services (`AuthService`, `ConnectivityService`, `ItineraryService`, `GuestStorageService`). Offline persistence combines `@ionic/storage-angular` (IndexedDB/LocalStorage backing store under keys `ne_guest_itineraries_v1` and `ne_is_guest_mode_v1`) with direct `window.localStorage` tokens (`ne_access_token`, `ne_refresh_token`, `ne_user_profile`, `ne_custom_api_url`).

3. Major Implementation Concerns & Code-Level Findings

1. **Duplicate FastAPI Route Handler & Dead Code in backend/app/api/v1/itineraries.py (Lines 365–413 vs. Lines 559–607):** The endpoint `@router.patch("/{itinerary_id}/days/{day_id}/items/{item_id}", response_model=ItineraryOut)` with function name `def update_itinerary_item(...)` is defined twice in the same file. FastAPI routes incoming requests to the first matching route in `router.routes` (Lines 365–413), which omits explicit `day_id not in valid_day_ids` validation and fails to update `item.updated_at` and `itinerary.updated_at`. The second definition (Lines 559–607), which contains the timestamp updates and day-membership check, is unreachable via HTTP routing.
2. **Triple Duplication of Offline POI Datasets:** Curated fallback places are hardcoded independently in three separate TypeScript files:
   - Project/src/app/tabs/home/home.page.ts (Lines 10–227: `DESTINATIONS` array)
   - Project/src/app/core/services/place.service.ts (Lines 41–403: `curatedOfflinePlaces` array)

- Project/src/app/core/services/itinerary.service.ts (Lines 592–910: `curatedPool` and `dayClusterPresets` arrays)

This violates DRY principles, bloats bundle size, and causes subtle data mismatches when offline vs. online.

3. **Deceptive/Stubbed Auth UI Handlers in login.page.ts (Lines 113–138) and signup.page.ts (Lines 158–174):**
   - `onForgotPassword()` (Lines 131–138 of `login.page.ts`) performs zero email validation and zero API calls, yet displays a toast stating: *"Password reset link will be sent to your registered email address."* No password-reset endpoint exists in `backend/app/api/v1/auth.py`.
   - `onGoogleLogin()` and `onGoogleSignup()` do not invoke Google Identity Services or `POST /api/v1/auth/oauth/google`; they only display a toast: *"Google Sign-In is configured for production credentials."*
   - `onAppleLogin()` on the Login screen (Lines 122–129 of `login.page.ts`) contains a copy-paste message typo: *"Apple Sign-Up is disabled in this pilot build."*
4. **Chronological Time-Sorting Flaw in Suggested Itinerary Output (suggested-wizard.page.ts & itinerary_generator.py):** During live testing of the Suggested Wizard, Day 1 ordered an "Overnight Stay" (*La Casa Prime Cabins*, 22 hrs) as Stop #2 between a 12:00 PM–2:00 PM cafe stop and a 2:00 PM–5:00 PM museum stop, and Day 2 placed a 10:30 AM stop (*Casa Batik*) before a 09:00 AM stop (*Farm Villaronte*).
5. **Incorrect Category Classification in Nearby Places Helper (place.service.ts, Lines 557–607):** In the Suggested Wizard's "Nearby" tab under `Stores`, cultural/museum attraction *Casa Batik* is surfaced as a store due to overly broad fallback bucketing in `getNearbyPlaces()`.
6. **Repository Bloat from Committed IDE Counter Artifacts:** The directory `.VSCodeCounter/2026-10-31_21-29-44/` is tracked in Git version control instead of being ignored via `.gitignore`.

C. Website Analysis (Phase 2 Live Inspection)

1. Visual Interface, Layout & Component Consistency

The live GitHub Pages deployment (https://edselrenz.github.io/Itinerary-App/) presents a cohesive, mobile-first visual design tailored to Philippine regional travel. Key visual observations from live inspection include:

- **Splash & Welcome Onboarding (`/welcome`, `/guest`):** Upon initial load, an animated splash overlay (`.splash-screen` in app.component.html) displays the Nueva Ecija Travel Itinerary emblem and auto-dismisses after 2.8 seconds (or immediately on click/swipe > 25px). The underlying Welcome screen renders the brand header *"Trip mo, trip ko"* and clear action hierarchy (`LOG IN`, `SIGN UP`, `Continue as Guest`).
- **Offline Status Banner Refactoring (Commit `06cefc5`):** When the backend tunnel is unreachable or paused, the top status banner renders a non-intrusive slate-900 glassmorphic bar (`role="status"`, `aria-live="polite"`) stating *"Offline Mode — Cloud sync paused. Your saved trips and curated destinations remain accessible locally"* with `Retry` and `Dismiss (X)` buttons. The legacy developer-facing `[Set Server] window.prompt()` button was removed in commit `06cefc5`.

- **Card & Image Rendering:** Destination cards on `/tabs/home` and the Place Detail modal render local WebP/JPG assets from `assets/images/places/` cleanly with category badges, price ranges (or explicit *"Cost not provided"* labels), and municipality subtitles.

2. Navigation & SPA Routing on GitHub Pages

- **Bottom Tab Bar (tabs.page.html):** Four bottom navigation tabs are present: `Home` (`/tabs/home`), `Map` (`/tabs/map`), `History/Saved` (`/tabs/history`), and `Profile` (`/tabs/profile`). Notice a minor labeling inconsistency: the route path and icon (`time-outline`) are named `history`, whereas the rendered page header in history.page.html (Line 5) is titled **"Saved Itineraries"**.
- **Direct Deep-Linking & Browser Refresh:** Because .github/workflows/deploy.yml copies `Project/www/index.html` to `Project/www/404.html` during build, refreshing the browser or deep-linking directly to sub-routes (such as `/Itinerary-App/tabs/home`, `/Itinerary-App/tabs/map`, or `/Itinerary-App/custom-builder`) loads the Angular router properly without breaking application state.

3. Responsive Behavior

Across desktop (1280×1024) and mobile-constrained viewport layouts, the Ionic grid and flexbox containers adapt cleanly without horizontal viewport overflow. Touch targets on filter chips, bottom tabs, and wizard step buttons maintain adequate spacing, and Leaflet map canvas resizing triggers `map.invalidateSize()` inside `ionViewDidEnter()` in map.page.ts to prevent gray unrendered tile strips.

D. Functional Quality Assurance (QA) & Test Execution Results (Phases 3 & 6)

Thirty structured test cases covering positive, negative, boundary, integration, navigation, UI, regression, responsive, and error-handling scenarios were designed against the actual repository features and executed against the deployed application (https://edselrenz.github.io/Itinerary-App/).

1. Test Execution Summary

Total Tests	Passed (PASS)	Failed (FAIL)	Partially Passed (PARTIAL)	Not Tested (NOT TESTED)
30	**21 (70.0%)**	**4 (13.3%)**	**3 (10.0%)**	**2 (6.7%)**

2. Complete Structured QA Test Suite & Execution Log

Test ID & Module	Test Scenario, Preconditions & Test Steps	Expected Result, Actual Result & Evidence	Status & Severity
TC-01 Splash & Routing	**Scenario:** Root URL load, splash screen dismissal, and redirect to `/welcome` (Navigation / UI) **Preconditions & Steps:** Open https://edselrenz.github.io/Itinerary-App/; wait 2.8s or click splash overlay.	**Expected:** App redirects to `/welcome`; splash fades out cleanly revealing hero title *"Trip mo, trip ko"*. **Actual & Evidence:** Redirected to `/welcome` with HTTP 200; splash overlay dismissed cleanly (`app.component.ts` lines 31–73).	**PASS** Low

Test ID & Module	Test Scenario, Preconditions & Test Steps	Expected Result, Actual Result & Evidence	Status & Severity
TC-02 Guest Onboarding	**Scenario:** Continue as Guest flow from Welcome to Home tab (Positive / Integration) **Preconditions & Steps:** On `/welcome`, click **Continue as Guest**, then proceed through `/guest` to `/tabs/home`.	**Expected:** Sets `ne_is_guest_mode_v1 = true`, clears auth tokens, and loads `/tabs/home`. **Actual & Evidence:** Transitioned to `/tabs/home`; Guest state persisted in storage (`auth.service.ts` lines 114–118).	**PASS** High
TC-03 Guest Persistence	**Scenario:** Browser hard refresh on deep-linked route `/tabs/home` (Regression / Storage) **Preconditions & Steps:** While in Guest Mode on `/tabs/home`, trigger browser reload.	**Expected:** `404.html` SPA fallback serves Angular bundle and retains Guest session without kicking user to `/welcome`. **Actual & Evidence:** Page reloaded on `/tabs/home` with guest session and saved trips intact.	**PASS** High
TC-04 Backend Connectivity	**Scenario:** Automatic health check against configured ngrok backend tunnel (Integration) **Preconditions & Steps:** Load app with commit `06cefc5` (`environment.prod.ts` pointing to `ngrok-free.dev/api/v1`).	**Expected:** `ConnectivityService.checkHostHealth()` sends GET `/api/v1/health` with `ngrok-skip-browser-warning: true`. **Actual & Evidence:** Initial load succeeded with no offline banner; 9 Nueva Ecija places loaded (`auth.interceptor.ts`, `connectivity.service.ts`).	**PASS** High
TC-05 Offline Banner	**Scenario:** Offline Mode banner retry and manual dismissal (Error-Handling / UI) **Preconditions & Steps:** When Offline Mode banner appears, click **Retry** and then click **Dismiss (X)**.	**Expected:** Retry shows crescent spinner while polling `/health`; Dismiss hides banner and sets `isDismissedManually = true`. **Actual & Evidence:** Retry button disabled during check with spinner; clicking `X` cleanly hid the banner (`app.component.html` lines 3–43).	**PASS** Medium
TC-06 Home Search (Positive)	**Scenario:** Search destinations by valid municipality name (Positive) **Preconditions & Steps:** On `/tabs/home`, enter `"Cabanatuan"` and then `"Gapan"` in search bar.	**Expected:** Displays matching places in Cabanatuan City (*Casa Batik*) and Gapan City (*Tagpuan sa Lumang Gapan*). **Actual & Evidence:** Returned *Casa Batik* for `"Cabanatuan"` and *Tagpuan sa Lumang Gapan* for `"Gapan"` (`home.page.ts` lines 310–332).	**PASS** Medium
TC-07 Home Search (Negative)	**Scenario:** Search destinations with non-existent string (Negative / Boundary) **Preconditions & Steps:** On `/tabs/home`, enter `"xyz123nonexistent"` into search bar.	**Expected:** Displays zero search matches cleanly without runtime exception or broken layout. **Actual & Evidence:** Returned 0 results; Search Results container remained cleanly empty/hidden without console errors.	**PASS** Low

Test ID & Module	Test Scenario, Preconditions & Test Steps	Expected Result, Actual Result & Evidence	Status & Severity
TC-08 Home Category Filtering	**Scenario:** Filter destinations by category pills and All Categories modal (UI / Functional) **Preconditions & Steps:** On `/tabs/home`, click `Nature & Parks`, `Restaurants`, `Churches & Faith`, and `All Places`.	**Expected:** Destination list updates immediately to show only places matching the selected category key. **Actual & Evidence:** Category pills updated active state and filtered displayed destination cards (`home.page.ts` lines 334–358).	**PASS** Medium
TC-09 Place Detail Modal	**Scenario:** Open and inspect Place Detail modal from Home card (UI / Integration) **Preconditions & Steps:** Click *Rohi Paperflower Cafe* and *Tagpuan sa Lumang Gapan* on `/tabs/home`, then click X.	**Expected:** Modal opens showing photo, badge, municipality, description, opening hours, price range, and route buttons; X closes modal. **Actual & Evidence:** All fields rendered accurately (`10:00 - 21:00`, `₱150 - ₱350`, `"Open in Google Maps"`, `"PLAN ROUTE HERE"`).	**PASS** Medium
TC-10 Map Tab Rendering	**Scenario:** Interactive Leaflet map load and custom marker rendering (Integration / UI) **Preconditions & Steps:** Click **Map** tab (`/tabs/map`) from bottom navigation bar.	**Expected:** Leaflet map initializes centered on Nueva Ecija with custom HTML pin markers (`L.divIcon`). **Actual & Evidence:** Map tiles and custom markers rendered properly without gray tile gaps (`map.page.ts` lines 162–215, 324–388).	**PASS** High
TC-11 Map Filtering & Selection	**Scenario:** Filter map markers by category pill and click marker pin (Functional) **Preconditions & Steps:** On `/tabs/map`, click `Nature & Parks` pill, then click a rendered marker pin.	**Expected:** Pill highlights in blue, non-matching pins are cleared, and clicking a pin opens the bottom place detail card. **Actual & Evidence:** Markers filtered dynamically and clicking a pin opened the bottom detail card (`map.page.ts` lines 280–414).	**PASS** Medium
TC-12 Suggested Wizard (Positive)	**Scenario:** Generate 2-day suggested itinerary with preferences and save (Positive / E2E) **Preconditions & Steps:** Open `/suggested-wizard`, select Nueva Ecija, Nov 2–3 2026 (2D/1N), budget ₱2,500/day, tags `Food Trip`, `Culture & History`, `Resort`; click **Find Itinerary** then **Accept & Save**.	**Expected:** Generates multi-day itinerary with stops, durations, and costs; saving persists trip and shows confirmation toast. **Actual & Evidence:** Generated *"Suggested Nueva Ecija Trip"* (Day 1 Bongabon 4 stops, Day 2 Cabanatuan 3 stops), saved with toast confirmation.	**PASS** High
TC-13 Suggested Wizard (Chronology)	**Scenario:** Verify chronological stop ordering and accommodation placement in generated plan (Functional / Logic) **Preconditions & Steps:** Inspect stop times and sequence on Day 1 and Day 2 of generated *"Suggested Nueva Ecija Trip"*.	**Expected:** Stops within each day should be ordered chronologically (morning to evening), with overnight stay at end of day. **Actual & Evidence:** Day 1 placed 22-hr Overnight Stay (*La Casa Prime Cabins*) as Stop #2 before 2:00 PM museum; Day 2 placed `10:30 AM` stop before `09:00 AM` stop.	**FAIL** Medium (DEF-03)

Test ID & Module	Test Scenario, Preconditions & Test Steps	Expected Result, Actual Result & Evidence	Status & Severity
TC-14 Suggested Wizard (Boundary)	**Scenario:** Budget input validation for zero and negative numbers (Boundary / Negative) **Preconditions & Steps:** In Step 2 of `/suggested-wizard`, enter `0` or `-500` into daily budget input.	**Expected:** Form control marks budget invalid and prevents proceeding with zero/negative budget. **Actual & Evidence:** Entering `0` triggered validation constraint preventing submission until valid budget (₱2,500) was set.	**PASS** Medium
TC-15 Suggested Wizard (Nearby Tab)	**Scenario:** Switch to "Nearby" view and inspect `Restaurants`, `Hospital & Pharmacies`, and `Stores` tabs (Functional) **Preconditions & Steps:** On Step 3 of Suggested Wizard, toggle `Nearby` view and click `Stores` tab.	**Expected:** Each tab lists only amenities matching that category in the active day's municipality. **Actual & Evidence:** Toggling worked, but `Stores` tab listed *Casa Batik* (a Culture & Museum attraction) as a store (`place.service.ts` lines 557–607).	**PARTIAL** Low (DEF-06)
TC-16 Custom Builder (Negative)	**Scenario:** Validate Trip Title minimum length and required fields on Step 1 (Negative / Boundary) **Preconditions & Steps:** Open `/custom-builder`, leave title empty or enter 2-character string `"QA"`.	**Expected:** Enforces `Validators.required` and `Validators.minLength(3)`, blocking progression. **Actual & Evidence:** 2-character title `"QA"` triggered validation error; entering `"QA Live Backend Test Trip"` enabled progression (`custom-builder.page.ts`).	**PASS** Medium
TC-17 Custom Builder (Positive CRUD)	**Scenario:** Create custom itinerary, add stops, reorder stops, verify budget calculation, and save (Positive / E2E) **Preconditions & Steps:** In `/custom-builder`, add *Rohi Paperflower Cafe* and *Museo de Bongabon* to Day 1, move *Museo de Bongabon* to Stop #1, click **SAVE**.	**Expected:** Stop order updates, budget summary shows `₱150 - ₱350 (+ 1 unpriced)`, and saving persists trip and navigates home. **Actual & Evidence:** Reordering, budget calculation, and save toast (*'Itinerary "QA Live Backend Test Trip" saved successfully!'*) worked as expected.	**PASS** High
TC-18 Saved / History Tab	**Scenario:** Verify saved Suggested and Custom trips appear in `/tabs/history` and open detail view (Integration) **Preconditions & Steps:** Click 3rd bottom tab (`/tabs/history`), verify list, and click a trip card.	**Expected:** Displays both *"QA Live Backend Test Trip"* and *"Suggested Nueva Ecija Trip"*; clicking opens `/itinerary/:id`. **Actual & Evidence:** Both trips rendered with correct badges, budgets, day/stop counts, and opened detail view (`history.page.ts` lines 79–83).	**PASS** High
TC-19 Itinerary Deletion	**Scenario:** Delete a locally saved guest itinerary from Saved Itineraries tab (Functional) **Preconditions & Steps:** On `/tabs/history`, click trash icon on a saved card and confirm deletion alert.	**Expected:** Confirmation dialog appears; confirming removes itinerary from storage and updates UI list. **Actual & Evidence:** `deleteItinerary()` (`history.page.ts` lines 85–123) prompts confirmation via `AlertController` and removes trip from storage.	**PASS** Medium

Test ID & Module	Test Scenario, Preconditions & Test Steps	Expected Result, Actual Result & Evidence	Status & Severity
TC-20 Login Form Validation	**Scenario:** Empty fields and malformed email validation on `/login` (Negative) **Preconditions & Steps:** Navigate to `/login`, enter invalid email `"notanemail"` or leave password blank.	**Expected:** Login submit button remains disabled (`[disabled]="loginForm.invalid"`). **Actual & Evidence:** Submit button remained disabled with `opacity: 0.5` until valid email and password were provided (`login.page.ts` lines 45–66).	**PASS** Medium
TC-21 Signup Form Validation	**Scenario:** Password minimum length (<8 chars) and password mismatch check on `/signup` (Negative / Boundary) **Preconditions & Steps:** Navigate to `/signup`, enter password `"short"` or mismatched `confirmPassword`.	**Expected:** `Validators.minLength(8)` and `passwordMatchValidator` keep Sign Up button disabled. **Actual & Evidence:** Form remained invalid and prevented submission until passwords matched and met 8-char minimum (`signup.page.ts` lines 62–93).	**PASS** Medium
TC-22 Forgot Password Action	**Scenario:** Click "Forgot password?" link on `/login` with empty or populated email (Functional / Error-Handling) **Preconditions & Steps:** On `/login`, leave email blank and click **Forgot password?**.	**Expected:** Should require a valid email address and invoke a backend password-reset endpoint (or clearly state feature is unavailable). **Actual & Evidence:** Displays deceptive toast *"Password reset link will be sent to your registered email address"* with zero validation and zero network call (`login.page.ts` lines 131–138).	**FAIL** High (DEF-01)
TC-23 Google OAuth Buttons	**Scenario:** Click Google Sign-In on `/login` and Google Sign-Up on `/signup` (Functional / Integration) **Preconditions & Steps:** On `/login` and `/signup`, click the Google authentication button.	**Expected:** Should launch Google OAuth flow and call `POST /api/v1/auth/oauth/google`, or be disabled if unconfigured. **Actual & Evidence:** Does not invoke OAuth or backend; shows static toast *"Google Sign-In is configured for production credentials"* (`login.page.ts` lines 113–120).	**FAIL** Medium (DEF-02)
TC-24 Apple Sign-In Button on Login	**Scenario:** Inspect Apple Sign-In handler and message on `/login` vs `/signup` (UI / Copy) **Preconditions & Steps:** Inspect `onAppleLogin()` in `login.page.ts` (Lines 122–129).	**Expected:** Toast/caption on Login screen should say *"Apple Sign-In is disabled..."*. **Actual & Evidence:** `onAppleLogin()` on the Login page says *"Apple Sign-Up is disabled in this pilot build."* (copy-paste typo).	**PARTIAL** Low (DEF-08)
TC-25 Backend Stop PATCH Route	**Scenario:** Partial update of an itinerary stop via `PATCH /api/v1/itineraries/{id}/days/{day_id}/items/{item_id}` (Backend Integration) **Preconditions & Steps:** Inspect route table in `backend/app/api/v1/itineraries.py` (Lines 365 and 559).	**Expected:** Single authoritative route handler validates `day_id` membership and updates `updated_at` timestamps. **Actual & Evidence:** Duplicate route registration at Line 365 shadows Line 559; requests hit Line 365 which skips `updated_at` updates.	**FAIL** Medium (DEF-04)

Test ID & Module	Test Scenario, Preconditions & Test Steps	Expected Result, Actual Result & Evidence	Status & Severity
TC-26 Profile Tab (Guest State)	**Scenario:** Inspect Profile tab UI and CTAs in Guest mode (UI / Navigation) **Preconditions & Steps:** Navigate to `/tabs/profile` while in Guest Mode.	**Expected:** Displays Guest Traveler status, unsynced local trip notice, and working buttons for `LOG IN TO SYNC` and `CREATE ACCOUNT`. **Actual & Evidence:** Rendered Guest Traveler banner and working navigation buttons to `/login` and `/signup` (`profile.page.html` lines 64–94).	**PASS** Low
TC-27 Documentation Parity	**Scenario:** Verify `docs/KNOWN_ISSUES.md` against actual History and Profile tabs (QC / Documentation) **Preconditions & Steps:** Compare `docs/KNOWN_ISSUES.md` Section 1.2 against live `/tabs/history` and `/tabs/profile`.	**Expected:** Documentation accurately reflects implemented Saved Itineraries and Profile sync views. **Actual & Evidence:** `KNOWN_ISSUES.md` falsely states History displays a deferred placeholder screen, whereas `history.page.html` is a full Saved Itineraries view.	**PARTIAL** Low (DEF-07)
TC-28 Responsive Viewport Layout	**Scenario:** Verify layout adaptation across desktop and mobile viewport dimensions (Responsive) **Preconditions & Steps:** Inspect `/tabs/home`, `/tabs/map`, and `/custom-builder` across desktop and mobile widths.	**Expected:** No horizontal scrollbars, clipped buttons, or overlapping tab bars. **Actual & Evidence:** Layouts adapted cleanly without horizontal overflow; map canvas resized properly.	**PASS** Low
TC-29 Live Account Registration & DB Mutation	**Scenario:** Submit new user account registration to live production database (Integration) **Preconditions & Steps:** Not executed on live server per Operating Rule 11 (prohibiting unauthorized account creation on live services).	**Expected:** Creates user record in PostgreSQL and returns JWT `TokenOut` (`auth.py` lines 37–56). **Actual & Evidence:** Verified statically via `auth.py` and `signup.page.ts`; not executed against live DB per Rule 11.	**NOT TESTED** Medium
TC-30 Authenticated Guest-to-Cloud Migration	**Scenario:** Migrate local guest trips to PostgreSQL upon authenticated login (Integration) **Preconditions & Steps:** Requires live authenticated account session (blocked by Operating Rule 11).	**Expected:** `GuestMigrationService.migrateGuestData()` posts each local trip to `POST /api/v1/itineraries` and clears local storage. **Actual & Evidence:** Verified statically in `guest-migration.service.ts` (lines 27–108); not executed on live DB per Rule 11.	**NOT TESTED** Medium

3. Manual Verification Procedures for Untested Live-Mutation Cases (TC-29 & TC-30)

In strict compliance with Operating Rule 11 (*"Do not submit forms, create accounts, delete records, or perform destructive actions on real services without explicit authorization"*) and Rule 14, live database account creation (TC-29) and authenticated cloud migration (TC-30) were verified via source inspection rather than mutating the production PostgreSQL database. To execute these two tests locally against a disposable test database, run the following reproducible procedure:

1. Start the local PostgreSQL and FastAPI backend (`python scripts/setup_postgres.py && uvicorn app.main:app --reload --port 8000` inside `backend/`) and launch the Angular app (`npm --prefix Project start`).
2. **For TC-29 (Account Registration):** Navigate to `http://localhost:4200/signup`, fill in `First Name: Test`, `Last Name: Auditor`, `Email: qa.auditor@example.com`, `Password: AuditPass123!`, and `Confirm Password: AuditPass123!`, and click **Sign Up**. Verify in browser DevTools Network tab that `POST /api/v1/auth/register` returns `201 Created` with `access_token` and `refresh_token`, stores `ne_access_token` in `localStorage`, and redirects to `/tabs/home`.
3. **For TC-30 (Guest-to-Account Migration):** In an incognito window, click **Continue as Guest**, create and save a Custom Itinerary in `/custom-builder`, verify `ne_guest_itineraries_v1` is populated in IndexedDB/LocalStorage, then navigate to `/tabs/profile`, click **Log In to Sync**, and sign in with `qa.auditor@example.com`. Verify that `GuestMigrationService.migrateGuestData(true)` issues `POST /api/v1/itineraries`, displays the migration toast, and clears `ne_guest_itineraries_v1`.

E. Quality Control (QC) Evaluation (Phases 4 & 9)

QC Pillar	Rating	Technical Evaluation & Supporting Evidence
Functional Quality	Good (82/100)	Primary travel discovery, map filtering, suggested itinerary generation, and custom itinerary building work end-to-end both online and offline. Deducted for non-functional Forgot Password / Google OAuth UI stubs and non-chronological stop times in Suggested Wizard results.
Architecture & Structural Quality	Good (80/100)	Clean separation between Angular standalone pages, RxJS core services, and FastAPI v1 routers. However, embedding 300+ lines of static POI arrays inside three different TypeScript files (`home.page.ts`, `place.service.ts`, `itinerary.service.ts`) creates maintenance drift and violates single-responsibility boundaries.
Maintainability	Moderate (74/100)	Files such as `itinerary.service.ts` (1,007 lines), `custom-builder.page.ts` (1,003 lines), and `itineraries.py` (706 lines, containing a duplicate route handler at lines 365 and 559) exceed reasonable module size thresholds and should be decomposed into smaller domain modules.
Reliability & Fault Tolerance	Strong (90/100)	Every HTTP read/write path in `PlaceService` and `ItineraryService` implements RxJS `catchError` fallbacks to local guest storage and curated datasets, preventing blank screens or unhandled crashes when the backend tunnel drops.
Performance Efficiency	Good (83/100)	Backend `places.py` sets `Cache-Control: public, max-age=3600, must-revalidate` and uses SQLAlchemy `joinedload(Place.municipality)` to avoid N+1 queries. Frontend uses lazy route loading (`loadComponent`), though bundling three duplicate POI datasets inflates client JS bundle size.

QC Pillar	Rating	Technical Evaluation & Supporting Evidence
Accessibility (WCAG 2.2)	Good (84/100)	The offline banner uses `role="status"`, `aria-live="polite"`, and explicit `aria-label` attributes on retry/dismiss buttons (app.component.html lines 4–41). However, the splash screen overlay (lines 49–83) uses a non-focusable `<div>` without keyboard `keydown.escape` dismissal or `tabindex="0"`, and custom Leaflet `L.divIcon` pins (map.page.ts) lack `role="button"` and `aria-label` attributes for screen readers.
Deployment Quality	Strong (88/100)	Automated GitHub Actions CI/CD workflow (.github/workflows/deploy.yml) builds production assets with `--base-href /Itinerary-App/`, creates `.nojekyll` and `404.html` for SPA deep-linking, and matches the live GitHub Pages deployment.

F. Defect Report (Phase 7 Classification)

Zero Critical defects were observed in the current build (the two historical BOLA/IDOR vulnerabilities `CR-001` and `CR-002` were confirmed remediated in backend/app/api/v1/itineraries.py via ownership predicates `Itinerary.user_id == current_user.id`). Ten verified defects across High (1), Medium (5), and Low (4) severities were established during code inspection and live QA testing:

Defect ID & Severity	Title & Affected Module	Description, Reproduction Steps & Expected vs. Actual Behavior	Root Cause, Evidence & Recommended Fix
DEF-01 HIGH	**Deceptive "Forgot Password?" Toast Without Validation or Backend Execution** `Project/src/app/features/auth/login/login.page.ts` (Lines 131–138) & `backend/app/api/v1/auth.py`	**Steps:** Navigate to `/login`, leave email empty (or enter invalid text), and click **Forgot password?**. **Expected:** Validate email input and call a password reset API endpoint (or explicitly inform the user that self-service password reset is not enabled in the pilot build). **Actual:** Immediately displays an Ionic Toast claiming *"Password reset link will be sent to your registered email address."* while making zero network requests.	**Root Cause:** `onForgotPassword()` in login.page.ts (lines 131–138) is a hardcoded UI toast stub with no backend route in `auth.py`. **Fix:** Either implement `POST /api/v1/auth/forgot-password` with email validation, or change the toast to clearly state that password recovery requires administrator assistance during the pilot phase.
DEF-02 MEDIUM	**Google Sign-In & Sign-Up Buttons Are Non-Functional Frontend Stubs** `Project/src/app/features/auth/login/login.page.ts` (Lines 113–120), `signup.page.ts` (Lines 158–165), `environment.prod.ts` (Line 23)	**Steps:** Navigate to `/login` or `/signup` and click the Google authentication button. **Expected:** Trigger Google Identity Services OAuth flow and pass the ID token to `POST /api/v1/auth/oauth/google`. **Actual:** Displays a static toast *"Google Sign-In is configured for production credentials."* and performs no authentication.	**Root Cause:** `onGoogleLogin()` and `onGoogleSignup()` only call `toastController.create(...)`, and `environment.prod.ts` has a dummy `googleClientId: 'prod-google-client-id.apps.googleusercontent.com'`. **Fix:** Wire Google Identity Services SDK to `POST /api/v1/auth/oauth/google` or disable/hide the button like Apple Sign-In until a real Client ID is provisioned.

Defect ID & Severity	Title & Affected Module	Description, Reproduction Steps & Expected vs. Actual Behavior	Root Cause, Evidence & Recommended Fix
DEF-03 MEDIUM	**Non-Chronological Stop Ordering in Suggested Itinerary Output** `backend/app/algorithms/itinerary_generator.py` & `Project/src/app/core/services/itinerary.service.ts` (Lines 586–999)	**Steps:** Generate a 2-day itinerary in `/suggested-wizard` for Nueva Ecija. **Expected:** Stops within each day are sorted chronologically from morning to evening, with overnight accommodation placed at the end of the day. **Actual:** Day 1 lists *La Casa Prime Cabins* (Overnight Stay, 22 hrs) as Stop #2 before afternoon stops; Day 2 lists *Casa Batik* (`10:30 AM`) before *Farm Villaronte* (`09:00 AM`).	**Root Cause:** Stop selection orders items by category/distance score without a final post-selection chronological sort by `planned_time` / accommodation category. **Fix:** Sort each day's selected items by parsed start time (placing `hotel_accommodation` / `resort_swimming` overnight stays last) before assigning `sort_order`.
DEF-04 MEDIUM	**Duplicate Route Handler `update_itinerary_item` Shadows Updated Implementation** `backend/app/api/v1/itineraries.py` (Lines 365–413 & Lines 559–607)	**Steps:** Send `PATCH /api/v1/itineraries/{itinerary_id}/days/{day_id}/items/{item_id}`. **Expected:** Executes the complete handler that validates `day_id` membership and updates `item.updated_at` and `itinerary.updated_at`. **Actual:** FastAPI matches the first handler at Line 365, which omits `updated_at` timestamp updates; the second handler at Line 559 is unreachable dead code.	**Root Cause:** During Phase 13 remediation (`CR-009`), a second `@router.patch("/{itinerary_id}/days/{day_id}/items/{item_id}")` function was appended at Line 559 without removing the earlier function at Line 365. **Fix:** Delete the first definition (Lines 365–413) and keep only the complete implementation (Lines 559–607).
DEF-05 MEDIUM	**Triple Duplication of Offline POI Datasets Across Frontend Modules** `home.page.ts` (Lines 10–227), `place.service.ts` (Lines 41–403), `itinerary.service.ts` (Lines 592–910)	**Steps:** Inspect offline fallback datasets in `Project/src/app/`. **Expected:** A single canonical offline POI data module shared across Home, PlaceService, and ItineraryService. **Actual:** Three separate hardcoded arrays exist with different schemas, item counts, and cost strings.	**Root Cause:** Incremental offline fallback additions across different sprints embedded separate static arrays in each service/component. **Fix:** Consolidate all offline POI records and cluster presets into a single `Project/src/app/core/data/offline-places.data.ts` module.
DEF-06 LOW	**Nearby Places Helper Misclassifies Museum Attraction Under "Stores"** `Project/src/app/core/services/place.service.ts` (Lines 557–607)	**Steps:** In `/suggested-wizard` Step 3, switch to **Nearby** view and click **Stores**. **Expected:** Displays retail/shopping/convenience stores. **Actual:** Lists *Casa Batik* (a Culture & Museum attraction) under `Stores`.	**Root Cause:** `getNearbyPlaces()` in place.service.ts maps non-restaurant/non-hospital fallback items into the `store` bucket. **Fix:** Filter strictly by retail/shopping category tags when populating the `store` category.

Defect ID & Severity	Title & Affected Module	Description, Reproduction Steps & Expected vs. Actual Behavior	Root Cause, Evidence & Recommended Fix
DEF-07 **LOW**	**Outdated Documentation in `docs/KNOWN_ISSUES.md` Regarding History & Profile Tabs** `docs/KNOWN_ISSUES.md` (Section 1.2) vs. `history.page.html` & `profile.page.html`	**Steps:** Compare `docs/KNOWN_ISSUES.md` Section 1.2 with the deployed `/tabs/history` and `/tabs/profile` views. **Expected:** Documentation matches implemented features. **Actual:** `KNOWN_ISSUES.md` claims History displays a placeholder *"Planned for a later release"*, whereas `history.page.html` is a fully implemented Saved Itineraries manager.	**Root Cause:** `KNOWN_ISSUES.md` was not updated after `history.page.html` and `profile.page.html` were implemented. **Fix:** Update `docs/KNOWN_ISSUES.md` Section 1.2 to reflect the implemented Saved Itineraries and Profile sync screens.
DEF-08 **LOW**	**Copy-Paste Toast Typo in `onAppleLogin()` on Login Screen** `Project/src/app/features/auth/login/login.page.ts` (Lines 122–129)	**Steps:** Inspect `onAppleLogin()` in `login.page.ts`. **Expected:** Message refers to *"Apple Sign-In"*. **Actual:** Message reads *"Apple Sign-Up is disabled in this pilot build."*	**Root Cause:** Copied handler body from `signup.page.ts` into `login.page.ts`. **Fix:** Change toast message in `login.page.ts` Line 125 to *"Apple Sign-In is disabled in this pilot build."*
DEF-09 **LOW**	**Accessibility Gaps on Splash Overlay and Custom Leaflet Map Pins** `Project/src/app/app.component.html` (Lines 49–83) & `map.page.ts` (Lines 324–388)	**Steps:** Navigate via keyboard (`Tab` / `Escape`) on splash screen and map markers. **Expected:** Interactive elements are keyboard-focusable and announce accessible names. **Actual:** Splash `<div>` lacks `tabindex` and `keydown.escape` listener; Leaflet `L.divIcon` pins use plain `<div>` markup without `role="button"` or `aria-label`.	**Root Cause:** Click/touch handlers were bound to generic `<div>` containers without keyboard/ARIA attributes. **Fix:** Add `tabindex="0"`, `role="button"`, and `aria-label="${this.escapeHtml(place.name)}"` to custom marker templates and support `Escape` key dismissal on the splash screen.
DEF-10 **LOW**	**Tracked IDE Counter Directory & Non-Conventional Commit Message in Git History** `.VSCodeCounter/2026-10-31_21-29-44/` & Commit `06cefc5`	**Steps:** Inspect repository root tree and `git log -n 1`. **Expected:** IDE extension artifacts excluded via `.gitignore`; commits follow conventional commit format. **Actual:** `.VSCodeCounter/` is committed to `main`, and commit `06cefc5` uses message `"tinanggalna nga niii"`.	**Root Cause:** Missing `.VSCodeCounter/` rule in root `.gitignore`. **Fix:** Add `.VSCodeCounter/` to `.gitignore` and remove cached files via `git rm -r --cached .VSCodeCounter`.

G. Security Review (Phase 5 Source Code Audit)

1. Confirmed Security Controls & Remediated Vulnerabilities

- **BOLA / IDOR Protection Verified:** All itinerary read, update, stop addition, stop deletion, and stop reordering endpoints in backend/app/api/v1/itineraries.py scope queries via `get_eager_itinerary_stmt(itinerary_id, current_user.id)` or explicit `Itinerary.user_id == current_user.id` filters.

- **SQL Wildcard & Injection Safety:** All database queries use SQLAlchemy 2.0 parameterized constructs (`select(Place).where(...)`), and search strings in backend/app/api/v1/places.py (Line 108) sanitize `%` and _ wildcards via `escape_like(q.strip())`.
- **Rate Limiting & Password Hashing:** Authentication endpoints in backend/app/api/v1/auth.py enforce SlowAPI rate limits (`5/minute` on `/register` and `/oauth/google`, `10/minute` on `/login` and `/refresh`), and passwords are hashed via bcrypt in backend/app/services/auth_service.py.
- **Removal of UI-Exposed Server Override Prompt (Commit `06cefc5`):** In commit `06cefc5`, the developer removed the `[Set Server]` button and `configureApiUrl()` prompt from `AppComponent`, eliminating the UI vector where a user could be socially engineered into pasting a malicious server URL that exfiltrates their Bearer JWT token via `authInterceptor`.

2. Potential Security Concerns & Hardening Recommendations

ID & Severity	Vulnerability / Security Concern & Exact Location	User Impact & Recommended Remediation
SEC-01 Medium	**Residual Unallowlisted `ne_custom_api_url` Override in `environment.prod.ts` & `auth.interceptor.ts`** Location: Project/src/environments/environment.prod.ts (Lines 1–16) & connectivity.service.ts (Lines 59–93)	**Impact:** Although the UI button was removed in commit `06cefc5`, `environment.prod.ts` still reads `localStorage.getItem('ne_custom_api_url')` on every API request and `authInterceptor` attaches `Authorization: Bearer <token>` to whatever URL is returned by `environment.apiBaseUrl`. Any XSS or shared-origin script setting `ne_custom_api_url` to an arbitrary `https://` host would exfiltrate JWTs. **Remediation:** Restrict `resolveApiBaseUrl()` in `environment.prod.ts` to an explicit hostname allowlist (`localhost`, `127.0.0.1`, `anybody-cube-surprise.ngrok-free.dev`).
SEC-02 Low	**JWT Access & Refresh Tokens Stored in `window.localStorage`** Location: Project/src/app/core/services/auth.service.ts (Lines 135–147)	**Impact:** Storing long-lived refresh tokens (`ne_refresh_token`) in `localStorage` makes them accessible to JavaScript if a future XSS vulnerability is introduced. **Remediation:** For web deployments, prefer `HttpOnly; Secure; SameSite=Strict` cookies for refresh tokens and keep short-lived access tokens in memory; on Capacitor mobile builds, use encrypted native keychains.
SEC-03 Low	**Default Development JWT Secret & DB Password Fallbacks in `backend/app/core/config.py`** Location: backend/app/core/config.py (Lines 11–16)	**Impact:** If the backend is launched in production without a `.env` file overriding `JWT_SECRET_KEY`, it defaults to the public string `"dev_only_replace_before_pilot_with_openssl_rand_hex_32"`, allowing attackers to forge valid HS256 JWTs. **Remediation:** Add a Pydantic `@model_validator(mode="after")` in `Settings` that raises a startup `ValueError` if `app_env != "development"` and `jwt_secret_key` equals the default placeholder.

H. Source Code vs. Deployed Website Comparison (Phase 8)

A direct comparison between the repository state on branch `main` (commit `06cefc5`) and the live GitHub Pages deployment at https://edselrenz.github.io/Itinerary-App/ confirmed the following:

- **Verified Consistent (Frontend Bundle):** The GitHub Actions workflow (.github/workflows/deploy.yml) automatically built and deployed commit `06cefc5` from `Project/`. The deployed site accurately reflects the removal of the `[Set Server]` button, the updated *"Offline Mode — Cloud sync paused"* banner text, the pre-baked ngrok endpoint (`https://anybody-cube-surprise.ngrok-free.dev/api/v1`), and the `404.html` SPA routing fallback.
- **Verified Architectural Split (Static Frontend vs. External Backend):** Because GitHub Pages is a static file host, only the compiled Angular frontend (`Project/www`) is hosted on `edselrenz.github.io`. The Python FastAPI backend (`backend/`) and PostgreSQL database run externally on the developer's host machine exposed via the persistent ngrok tunnel (`https://anybody-cube-surprise.ngrok-free.dev`). When the host tunnel is active, the live website operates in full client-server mode; when the host tunnel is stopped, the website operates in local offline fallback mode.

I. Recommended Improvements (Phase 10 Action Plan)

1. Immediate Fixes (Before External User Testing)

- **Remove Duplicate Route Handler in `itineraries.py` (DEF-04):** Delete Lines 365–413 in backend/app/api/v1/itineraries.py so that `PATCH /{itinerary_id}/days/{day_id}/items/{item_id}` executes the complete handler at Lines 559–607 and properly updates `updated_at` timestamps.
- **Remediate Deceptive Auth Stubs (DEF-01, DEF-02, DEF-08):** Replace the misleading *"Password reset link will be sent..."* toast in login.page.ts (Lines 131–138) and the stubbed Google Sign-In/Sign-Up buttons with either functional backend flows or explicit `[disabled]="true"` states with honest pilot-scope captions; fix the *"Apple Sign-Up"* copy-paste typo on Line 125 of `login.page.ts`.

2. High-Priority Improvements

- **Sort Suggested Itinerary Stops Chronologically (DEF-03):** Add a post-clustering sort step in itinerary_generator.py and itinerary.service.ts so daily stops are ordered from earliest morning arrival time to evening, placing overnight hotel/cabin stays at the end of each day.
- **Allowlist Custom API Domains & Enforce Production Secret Check (SEC-01, SEC-03):** Restrict `resolveApiBaseUrl()` in environment.prod.ts to trusted hostnames and enforce a startup check in backend/app/core/config.py rejecting default JWT secrets outside development.

3. Medium-Priority Improvements

- **Consolidate Offline POI Datasets (DEF-05, DEF-06):** Extract the duplicated place arrays from `home.page.ts`, `place.service.ts`, and `itinerary.service.ts` into a single typed data module, and fix `PlaceService.getNearbyPlaces()` (Lines 557–607) so cultural attractions like *Casa Batik* are not categorized as `Stores`.
- **Synchronize Documentation & Git Hygiene (DEF-07, DEF-10):** Update docs/KNOWN_ISSUES.md Section 1.2 to reflect the completed Saved Itineraries and Profile tabs, and remove `.VSCodeCounter/` from version control.

4. Long-Term Architectural Improvements

- **Cloud-Hosted Backend & Managed PostgreSQL:** Migrate the FastAPI backend and PostgreSQL database from a local workstation ngrok tunnel to a managed cloud runtime (e.g., Render, Railway, Fly.io, or Cloud Run with Neon/Supabase PostgreSQL) for 24/7 availability without host-machine dependency.
- **Accessibility & Automated CI Gateways (DEF-09):** Add keyboard/ARIA attributes to custom Leaflet markers and splash overlays, and integrate the existing Vitest unit tests (`npm run test:unit`) and Pytest suite (`pytest backend/tests/`) into .github/workflows/deploy.yml as pre-deployment gates.

J. Final Verdict

Verdict: NEEDS MINOR IMPROVEMENTS — Generally functional but has outstanding issues.

Reasoning: The Nueva Ecija Travel Itinerary Application demonstrates strong engineering thoughtfulness in its core travel-planning domain, particularly in its resilient offline-first fallback architecture, Haversine geographic clustering, BOLA/IDOR authorization checks, and automated GitHub Pages SPA deployment pipeline. Live interactive QA confirmed that Guest onboarding, destination search, category filtering, interactive Leaflet mapping, Suggested Itinerary generation, Custom Itinerary creation/reordering, and local persistence work reliably. However, the presence of deceptive non-functional authentication stubs (`onForgotPassword`, `onGoogleLogin`), a shadowed duplicate PATCH route handler in `backend/app/api/v1/itineraries.py`, non-chronological stop ordering in generated itineraries, and triple-duplicated offline datasets prevent an unconditional `ACCEPTABLE` rating until those targeted remediations are applied.